

National Textile University, Faisalabad



Department of Computer Science

Name	Minahal Fatima
Reg no	23-NTU-CS-1050
Section	BSCS 7 th -A
Subject	Machine Learning
Lab Manual	1
Submitted to	Miss Kianat

14/9/26

Date:

Monday.

Day:

Machine Learning Lab 1

Random Integer:-

- Seed
- read_csv
- Random (SD, Variance).

Using sklearn:-

Regression, classification dataset
will available

→ load_iris

iris.data (to see features)

iris.target (to see targets).

Data preprocessing.

→ dropna (to drop ^{all} values of rows or
columns
axis=0
axis=1)

→ fillna (to replace missing values with
mean/median).

Encoding (apply on target value).

↳ alphabets to numeric

→ manual
encoding
↳ for ordered
data

→ Label
↳ for unordered

Date: _____

Page: _____

outliers

↳ boxplot → to see outliers in a dataset.

↳ IQR → to manage outliers.

↳ log transformation. ↙

Feature scaling :-

If variation exist (means values in columns are differ).

→ StandardScaler() → Deep learning.

MinMaxScaler() → machine learning.

(0-1 arrangement).

Correlation (feature selection).

→ from already existing feature generate new ones → feature Engineering.

Dimension Reduction (PCA).

→ combine columns to reduce dimensions.

Date: _____ Imbalanced dataset. Day: _____

If data of two categories not present in equal amounts.

→ over sampling → SMOTE (g_1^{50}, g_2^5) (equal both)
→ under sampling → Random under sampling ().

to avoid biasness, for proper training.

train, test.
80% 20%

→ class weighting.